

FoundationPose x Threestudio setup

DISCLAIMER: It is assumed the user has already installed docker and set it with admin permissions

Step 0: Setup docker to run without sudo

```
Unset
sudo groupadd docker
sudo usermod -aG docker $USER
newgrp docker
docker ps
```

Step 1: Clone The Repository (if you haven't done so already)

```
Unset
https://github.com/JustinLungu/FoundationPose-BachelorThesis.git
cd FoundationPose-BachelorThesis
```

Step 2: Setup the virtual environment

First make sure you have all the required dependencies to run all pipelines in the project, including the population of the repository:

```
Unset
cd ..
conda env create -f environment.yml
#OR
pip install -r requirements.txt
```

Step 3: Populate the project completely

```
Unset
python populate_pipelines.py
```

To download all the necessary data, including the threestudio setup environment, such that all pipelines can be run successfully.

Step 4: Build the project (both FoundationPose and threestudio images)

```
Unset  
./build_project.sh
```

Note: It will take a while (about an hour) to setup everything properly.

Step 5: Run the Project

```
Unset  
./run_project.sh
```

Now we will open FoundationPose container.

Step 6: Build FoundationPose from inside the container

Since it's our first time running it, we will first need to build all necessary dependencies:

```
Unset  
bash build_all.sh
```

You do not need to repeat this step when running the FoundationPose container again.

Step 6: Test FoundationPose individually

If there is no demo_data folder make sure to checkout the FoundationPose individual setup document for reference:

```
Unset  
python run_demo.py
```

Step 6: Test Threestudio individually

If there is no demo_data folder make sure to checkout the FoundationPose individual setup document for reference:

```
Unset  
cd ..  
cd threestudio/docker  
./run_container.sh
```

It should create a “bunny on a stack of pancakes” in the outputs directory in threestudio.

Step 6: Test FoundationPose x Threestudio

Now we test that for the demo setup, if there is no object found, we will generate one using threestudio:

```
Unset  
cd ..  
cd ..  
cd Host_pose_estimation  
python main.py
```

If .obj exists then you will see FoundationPose predicting poses, otherwise, threestudio generating the object, then FoundationPose predicting poses.